

Action, Trajectories and Gaps: A Research Programme

navstokgap research working notes

Abstract

We organize the accepted action-scale results around the physical premises that supply positivity and the counterexamples a selection principle must exclude. Quantum reconstruction and interacting spectral control are the main research tracks. The programme identifies their next proof obligations and keeps dimensional action selection, calibrated observable access and uniform limits explicit. The maintained source is `research/PROGRAMME.md`; `research/STATE.md` supplies the single current priority list.

1 Research programme: quantum necessity and robust gap mechanisms

Historical (2026-09-16). Superseded by `AGENTS.md`, which removed the per-step bookkeeping this file prescribes. Kept as context; not instructions.

Version 72, 2026-09-16. The programme consolidates the accepted results into Classical action scales: obstructions, conditional bounds and quantum premises. The user's clarified 2026-09-14 priority is a mathematical necessity argument for the Newton/Galileo area-action obstruction; gap work is supporting. `STATE` owns the current ordered tasks; `STRATEGY` governs bounded choices and stopping decisions.

1.1 Aim and present conclusion

Derive the need for quantum structure with a positive universal action parameter from explicit physical premises, and develop a precise mechanism for a gap that survives the relevant limits. A construction admitting quantum inputs and a construction selecting those inputs answer different questions. Newtonian cut refinement remains the consistency test for proposed action mechanisms; the Millennium comparison supplies the field-theory target.

The accepted examples provide a substantial set of obstructions to simple classical selection arguments and several conditional positive bounds. The physical premise excluding their quiet, contracting or precisely observed countermodels remains open. Spectral work gives finite-system criteria, independent-product control and C124's interacting Ising gap uniform in finite volume at bounded coupling. Physical-operator identification and further uniform limits remain open.

This version integrates C129/B77: the initial reversible-generator restriction is proof-audited, with the later interaction-theorem steps still open. C128/B74 establishes that the exact finite observable repair of C127 is impossible for the admitted spin preparations. G04 retains its local Hermitian parent C126 alongside G03's interacting gap C124. No novelty is asserted. The ledger retains proof and literature status, including the limits of each source audit.

1.2 Quantum track: identify the excluded classical alternative

Q01's first milestone compares Hardy and Chiribella–D'Ariano–Perinotti through the exact premise that rules out a classical operational model. The bounded premise map and B67 audit are complete: Hardy requires continuous reversible pure-state connectivity; CDP requires pure extensions with essential uniqueness. Standard finite classical models fail these premises. Hardy's

moving-ball example points to the extra question of an exact finite operational description closed during transformations. Neither full reconstruction proof has been audited here. For each selected passage specify the state space, preparations, transformations, composition and measurement assumptions. Test the classical alternative in that same class and mark the step where action units would still need to enter.

K01 is the supporting fixed-time compatibility test. It asks whether local measurement descriptions extend to one system-plus-apparatus state and which assumption about context independence changes the answer. A primary Kochen–Specker comparison is a source task with explicit hypotheses. Merely hiding apparatus variables is not enough to exclude their classical model. I007 records the construction.

The B01 companions provide the existing Hardy/CDP source routes and limited reading coverage. Full reconstruction-proof acceptance remains future work. The historical source-to-model connections are preserved under their own tasks and can supply a specific premise or example.

1.2.1 Action selection obligations

The user’s central example is Galileo’s falling parabola compared with the inertial horizontal line. Its enclosed area A obeys $(3F/v)A = \tau\Delta E$, with $\Delta E = F^2\tau^2/(2m)$ the gained kinetic energy. The matched-endpoint chord lens is half that area; Kepler’s swept sectors are a different observable. The objective is a physical obstruction to indefinite refinement into distinguishable classical motion, connected to a positive universal action scale.

N01 makes a closed comparison starting with the falling parabola: an explicit return-force sequence reunites the arms with equal positions and momenta, and its exact quantum relative phase is $-2F^2\tau^3/(3m\hbar)$. With C006 it gives a positive resolution threshold at fixed copies and accuracy. A hard area cutoff does not follow from the deterministic energy gain alone; the quantum construction supplies $\hbar$. This is a supporting quantum benchmark. The intended innovation is a mathematical necessity argument from independently justified consistency premises that exclude zero-action refinement without supplying quantum kinematics. N02 tests one such proposed principle with an explicit cut state, composition rule and joint time/position limit. The N01 construction remains exploratory and supplies no necessity theorem.

Q14’s dimensional criteria consolidate the selection countertests: a floor shared by all admitted masses and preparations is a dimensionless multiple of an action-dimensional product of the fixed constants, and an admitted similarity that rescales the observable forces the floor to zero. The fixed constants of classical mechanics, electrodynamics and gravitation admit one mass-independent action, $k_e/c = e^2/(4\pi\epsilon_0 c) = \alpha\hbar$, and C052’s relativistic Kepler threshold attains it. The open selection premises are a fixed charge unit and the factor $1/\alpha$. The criteria remain exploratory. C130, accepted by user direction after the collapse audit, records that C052’s plunge threshold and circular energies at angular action $\hbar|\kappa|$ are the Dirac Coulomb thresholds and zero-radial-node levels, and that radial action quantization of C052 gives the Sommerfeld formula, which is the Dirac spectrum; Planck’s constant enters only as the unit of angular action at the plunge boundary.

Q12’s local receiver test derives an all-gate coincidence-product bound for independent monotone responses to fixed fractions of one classical pulse, and a sharp efficiency/balance trade-off. Fluctuating routing, shared readiness and outcome-based gate selection supply explicit countermodels when the corresponding premises are removed. This exploratory result limits a receiver class; its dimensionless statistics supply no action scale. Shared-preparation models require a separate test involving independently chosen local settings.

Q10’s thermal receiver test combines a dark retention interval with a response deadline. Its activated-rate model yields an exact barrier-change floor proportional to temperature and a logarithmic timing/error contrast. A fixed signal-to-work coupling is needed to turn that into an incident-energy bound. Kinetic gating exposes that missing premise; temperature and appara-

tus parameters remain. This closes arbitrary preloading only within the stated effective model, without ledger promotion.

Q08's mechanical interference test constructs an ideal lossless string network with two paths and matched work-recording receivers. Exact finite-pulse output energies contain the wave autocorrelation; long coherent pulses give cosine fringes. Attenuation preserves normalized fringes while energy and canonical action vanish. The canonical action-to-phase ratio depends on preparation and impedance; the traveling wave's spacetime Lagrangian action is zero. This exploratory result tests Q07's physical phase premise without ledger promotion. Localized event formation and its energy source remain a distinct receiver question.

Q06's reciprocal coupling test constructs conservative exchange with measured acceleration responses. Reciprocity determines relative inertias, subject to a network cycle condition; one common energy/action multiplier survives. This exploratory construction extends Q05's single-field normalization test to interacting sectors without promoting a ledger claim. It supplies relative calibration, while absolute normalization and the quantum phase role remain separate obligations.

C127's Hamiltonian test now separates microscopic reversibility from operational descent. Two zero-energy spin preparations share all retained first/cross moments but an allowed product test distinguishes them after the interaction. The finite-repair proof C128/B74 now rules out every finite observable enlargement on a nonzero time interval for all Borel preparations, including nonlinear updates of finite expectation vectors. Exact closure requires infinite observable dimension or a changed physical premise. This settles the named repair alternative; it selects neither quantum structure nor a universal action unit. Further hierarchy variants are parked while the missing physical premises are consolidated.

P06 now consolidates these results into a standalone spin manuscript, compiled PDF and PDF-inclusive bundle. The consistency review maps claims to existing audits. B76 supports a worked conceptual example: classical extensions and transformation-relative preparation equivalence supply established precedents; the explicit spin pair and finite-repair proof supply the model-specific presentation. The manuscript now distinguishes the initial quotient from the expanded experimental theory. Its action-selection premise remains a separate physical question.

Q04's topological field test constructs a conservative $O(3)$ sector with energy at least $4\pi\rho$, while an exact degree-one family of radius R has radius-crossing action $4\pi\rho R/c \rightarrow 0$. The static energy floor coexists with massless vacuum waves and translation zero modes. This exploratory test separates sector exclusion from a physical size or time scale; Q03's fueled attraction and Q04's geometric restriction have not supplied a universal action constant.

A candidate must discharge five gates in the action target: classical definition and units; exclusion of zero; convergence in a named topology; universality across the admitted masses and preparations; and identification with quantum phase or commutation structure. Define $h_\varepsilon(t)$ when using the proposed action-field route, keeping physical time t and resolution ε distinct. Observation duration Δ , volume and c^{-1} have their own limits.

The checkerboard calculation takes a supplied action parameter, amplitudes and measurement rule to a coherent Dirac limit. It is the Branch A comparison for a Branch B necessity argument. A state-space reconstruction still needs a mechanical dynamical identification of its dimensional scale; numerical calibration is a further empirical task.

1.3 Gap track: interactions and uniform control

Q01's orientation countermodel (C125/B69) makes the quantum-track boundary explicit: finite capacity and reversible local closure coexist with classical separable composition, failed purification and failed subspace equivalence. The B70 interaction audit now excludes a continuous reversible nonlocal extension preserving that minimal composite, using de la Torre et al.'s Theorem 1. The local ball and Hermitian coordinates already meet the local-qubit premise. Identical-copy consistency and operational ancilla/measurement/discard closure underlie the stronger quantum

reconstruction. The full theorem proof remains unaudited here; physical reversible descent and action normalization remain open. G04's Hermitian transfer (C126/B71) gives a positive three-site local parent with square-root Gibbs ground state and energy-unit gap $Ka[1 - \tanh(2b)]$. Both action constant K and the physical clock identification are supplied. This resolves the finite-volume operator construction while retaining the independent physical-dynamics premise.

G05's direct Hamiltonian works through an established cluster chain obtained by a depth-two controlled-Z circuit. It has three-site interactions, an entangled ground state and exact energy gap J uniform in chain length at fixed lattice spacing and coupling. Endpoint terms control ground-state uniqueness. This gives a direct quantum energy operator without a sampling-clock identification, with quantum kinematics and the action factor for physical time supplied. Robustness under a local noncommuting perturbation is a separate test; no new ledger claim is promoted.

G07's dictionary note answers the user's 2026-09-16 question of how $h > 0$ is analogous to the mass gap. The floor/unit theorem of Q14 applies verbatim to both observables: a positive finite floor is a dimensionless multiple of a unit formed from the fixed constants, and classical Yang–Mills in four dimensions and classical mechanics with a mass and a force both lack the unit. The note proves, with $\hbar$, g and m explicit, that Yang–Mills quantum mechanics (the zero-momentum sector of the torus theory) has a purely discrete spectrum and gap $\delta_1 \hbar^{4/3} g^{2/3} m^{-2/3}$ which vanishes when either the canonical or the Lie-algebra commutator is removed; that free and abelian fields inherit their gapless classical dispersion; that two-dimensional Yang–Mills on a circle has a gap $\frac{1}{2} g^2 \hbar^2 c L C_2$ diverging with the volume; and that scale-invariant quantum mechanics obtains its scale from the self-adjointness requirement. The finite-volume zero-mode gap $\delta_1 g^{2/3} \hbar c / L$ exhibits Jaffe–Witten's uniformity problem in a solved sector. Consequence: C124/C126 and G05 are supplied-scale gaps; a generated gap needs two scale-free non-commuting structures; the analogy covers the structure-to-scale stage of the necessity question and points N02 to a consistency-forced scale. Exploratory; sources in B78; no promotion.

G08 turns the analogy into an implication inside that model. Classical mechanics plus the postulate that no transverse oscillation has phase-space area below $h/2$ confines the valley motion to $|x| \leq 2\sqrt{m} E / (\hbar g)$, gives the finite phase-space volume with Simon's $E^{3/2} \ln E$ growth and reproduces the exact energy unit by Bohr–Sommerfeld quantization; the operator inequality of G07 is this floor enforced by the uncertainty inequality. The abelian model has no transverse oscillation and stays gapless with $h > 0$ assumed. Conversely any positive gap in the (m, g) model supplies the action unit $\Delta^{3/4} m^{1/2} g^{-1/2}$; the same equivalence holds for Yang–Mills in $d = 2, 3$ and fails at $d = 4$, where a mass gap with c admits no action unit. Exploratory; sources in B79; no promotion.

G01/G02 establish the finite reversible-generator estimate

$$\gamma \geq \frac{\alpha}{S}, \quad S = \sum_a \langle f_a, (-Q)^{-1} f_a \rangle,$$

where α is the lower observable-frame bound on centered states. One positive measured plateau can miss an arbitrarily slow mode; exact state identification can also become poorly conditioned. Independent product dynamics control mixed modes and give the minimum constituent gap when each constituent clock is retained (C041–C046).

G03 now supplies an explicit interacting estimate: the periodic zero-field Ising heat-bath chain has full gap $a[1 - \tanh(2b)]$ for every $N \geq 3$. Here a is the per-site refresh rate and $b \geq 0$ the dimensionless nearest-neighbour coupling. Hamming contraction controls all modes and magnetization attains the bound. Bounded coupling and a clock floor give a uniform finite-volume gap; finite range alone permits joint coupling/size closure, and a fixed total clock adds a $1/N$ slowdown. The C124 proof replaces tensorization without an extensive full-frame susceptibility sum. B68 records a bounded model/method audit, including unavailable Glauber formula images. Further interaction variants are parked until a named operator or limit dependency requires them. See also the susceptibility proof.

A Yang–Mills transfer additionally requires its physical quantum Hamiltonian, gauge-invariant observables, continuum and infinite-volume construction and quantum axioms. An auxiliary sampling generator’s relaxation rate changes under clock rescaling while its invariant law stays fixed. The comparison note and official-target digest preserve these distinctions. A finite-box decay estimate or a toy Hessian gap supplies only its own operator statement.

1.4 Consolidated evidence and retained supporting models

Mechanism tested	Accepted consequence	Premise still requiring selection
Continuous variation and scale contraction	C002/C056–C058 give zero action infima in specified classes	Restriction excluding the allowed variations or low excitation
Positive orbit cost	C060–C061 give a sharp force/speed-floor bound	Physical origin and universality of the excitation floor
Fluctuation and independent composition	C035–C036 give a common coefficient using a positive reference	Preparation equivalence and origin of reference fluctuations
Prepared apparatus and records	C068–C123 give recovery and ambiguity under explicit data restrictions	Irreducible, universal information/preparation constraints
Coherent paths	C039–C040 give continuum and measured-cut comparisons	Necessity of amplitudes, measurement rule and action parameter
Observable response and interaction	C041–C046 give gap criteria; C124 gives an interacting uniform finite-volume gap	Origin of coupling/clock control, physical operator identification and further limits

The consolidated paper supplies short arguments and links to the full proofs. The receiver/calibration compilation retains technical details, including R33’s open curvature test. R33 is parked: its next coefficient currently changes no selected quantum or gap premise. It can return for a named dependency, proof correction or an explicit user request.

A19’s peak-excitation estimate, A14’s renewal-preparation model, R02’s repaired finite-propagation drafts and M03’s spectral laboratory remain supporting work. They become selected work when their possible outcomes change a main premise decision. New solvable subcases do not automatically displace the main tracks.

1.5 Cut refinement and historical context

The cut-point target collects the existing consistency tests. Constant-force chord error closes under refinement; fixed Gaussian bridge laws preserve their supplied fluctuation parameter; velocity and receiver memory distinguish retained-state cuts from interventions. Joint path/action limits require their stated topology and preparation changes. These results constrain proposed mechanisms without selecting quantum weights.

Newton’s geometric arguments, the cone/arrow proposals, the Classical Scholia and the H10 static-composition readings remain source-backed research inputs. The bibliography bridge index routes a named task to a construction, premise or exact passage. H11 strengthens source witnesses when selected; it does not resume merely because an old source card calls it next. The supplied bibliography note remains intact, with its separate mathematical selection.

1.6 Publication development

The user has selected publication preparation. P05 extracts C050–C051 into Same collisions, different transport, a standalone teaching draft with a PDF-inclusive source bundle. B75 finds a close Brownian random-versus-regular preparation precedent; the exact Hamiltonian matched-rate comparison is unmatched only within the bounded search. No new theorem or novelty claim is introduced. The spin draft consolidates C125–C128, including the completed finite-repair obstruction. P07 completes accessibility and contribution edits for both drafts; the author decision sheet recommends the gas paper first and records the remaining author and destination choices. STATE owns ordering; journal submission remains a separate authorized action.

1.7 Acceptance, cost and consolidation

Use written proofs and bounded source review. No numerical or symbolic verification scripts may be created or run in Python or another language. Historical scripts and outputs remain records. Document/source/integrity Python is allowed. Delegation is selective and sequential under the protocol, with one Sol/Luna worker and explicit supported effort, never ultra.

Before each step, record its target obligation, the decision changed by its possible outcomes, its deliverable/stop and why existing results are insufficient. After two consecutive steps in a proof family, compare its continuation with the other main track. The coordinator makes that choice without routine user approval. Correct substantive errors when found and preserve explicit user scope. A completed local calculation need not spawn another calculation.

A new result receives written proof review and a bounded librarian audit. Consolidation reuses those audits with an evidence map; it does not imply a fresh prior-art search or automatic novelty. Keep STATE short and reconcile its queue with TASKS and the latest handoff. Dated milestones live in the state history and individual handoffs.

Run `make check`, `make papers` and `git diff --check`; inspect changed PDF pages and preserve unrelated worktree changes. Relevant status/results are committed and pushed under standing authorization. External submissions, correspondence and paid services need separate direction. Publication claims must reflect the accepted proofs and the actual literature coverage.